

Circuit Simulation Lab:21

Design of Counters

The purpose of this experiment is to introduce the design of LFSR Counters.
An LFSR is a shift-register with some XOR gates

CODE:

```
module lfsr_8bit(
input clk,
output reg [7:0] q
);

wire feedback;

assign feedback = q[7] ^ q[5] ^ q[4] ^ q[3];

initial
q = 8'b11111111;

always @(posedge clk)
begin
    q <= {q[6:0], feedback};
end

Endmodule
```

TESTBENCH

```
module tb_lfsr_8bit;

reg clk;
wire [7:0] q;

lfsr_8bit dut(
.clk(clk),
.q(q)
);

always #10 clk = ~clk;

initial begin

$dumpfile("lfsr.vcd");
```

```
$dumpvars(0,tb_lfsr_8bit);
```

```
clk = 0;
```

```
#500;
```

```
$finish;
```

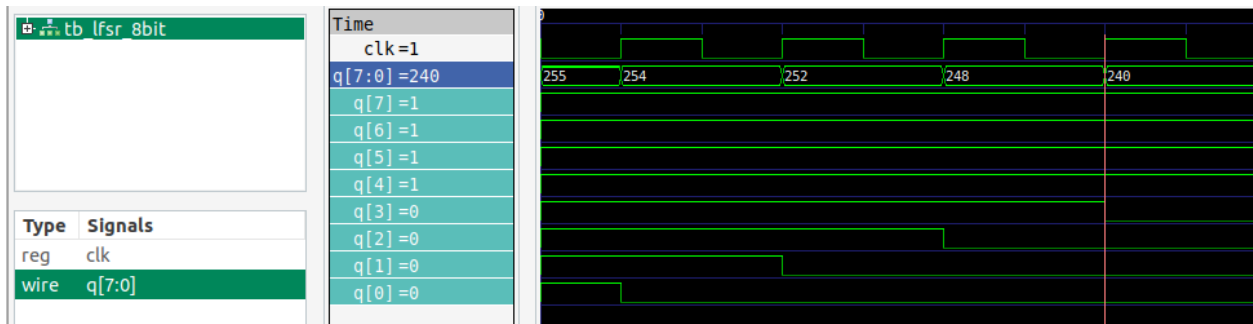
```
end
```

```
initial begin
```

```
$monitor("time=%0t q=%b",  
          $time,q);
```

```
end
```

```
endmodule
```



```
ll.v  
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_21$ iverilog -o out  
lfsr_8bit.v tb_lfsr_8bit.v  
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_21$ vvp out  
VCD info: dumpfile lfsr.vcd opened for output.  
time=0 q=11111111  
time=10 q=11111110  
time=30 q=11111100  
time=50 q=11111000  
time=70 q=11110000  
time=90 q=11100001  
time=110 q=11000010  
time=130 q=10000101  
time=150 q=00001011  
time=170 q=00010111  
time=190 q=00101111
```